

LIFELINE-RL: A BENCHMARK FOR TOPOLOGY-AWARE AND HISTORY-DEPENDENT SELF-PLAY

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ABSTRACT

We introduce LIFELINE-RL, a deterministic self-play benchmark combining a triangular lattice, player-specific logical connectivity, cascading removal, and history-dependent repetition. Complete side-five enumeration finds reachable latent topology differences but no difference in legal actions, transitions, exact values, or optimal actions; a standard-start side-six replay instead shows Superko preventing a repeatable six-transition loop. We compare a padded CNN, physical-edge Grid-GNN, and typed-edge Topology-GNN in 15 matched mixed-size runs over sides 5, 7, and 9. All three beat Random at every seen size. Against UCT-MCTS-16, seven of nine model-size cells have descriptive Wilson lower bounds above 0.5, one is borderline, and one is inconclusive. Topology-GNN outscores Grid-GNN in direct play at all three sizes, but each 200-game task repeats only two deterministic trajectories; the evidence does not support a global representation ranking. We release frozen protocols and replay-verified artifacts.

1 INTRODUCTION

Most board-game benchmarks expose local occupancy on a fixed lattice. LIFELINE-RL instead separates rendered occupancy from player-specific logical connectivity: moves can cut edges, remove disconnected components, and invoke a repetition rule whose legality depends on history. We ask which representations support self-play under these mechanisms.

Our contributions are:

1. a tested variable-size environment with exact full-state snapshots, legal masks, and replayable trajectories;
2. exhaustive and bounded diagnostics separating latent topology aliasing, decision-relevant aliasing, and Superko activation; and
3. a five-seed, compute-matched comparison of three representations with 30,600 replay-verified formal evaluation games.

The environment contribution does not depend on a positive model result. We claim neither non-Markov observations, relative sample efficiency, nor unseen-size generalization without the corresponding test.

2 THE LIFELINE-RL BENCHMARK

2.1 BOARD AND ACTIONS

A board of side length n contains

$$N(n) = \frac{n(n+1)}{2} \tag{1}$$

valid lattice points. A turn selects one lattice point or the pass action, for $N(n) + 1$ discrete actions. The main benchmark uses two players and canonicalizes the observation to the player to act.

Table 1: Pre-registered observation variants. The simulator always retains the exact full state.

Name	Exposed structure	Purpose
Grid	Occupancy and turn features	Flat/spatial baseline
GridGraph	Grid plus fixed lattice edges	Size-agnostic graph baseline
Topology	GridGraph plus typed logical edges	Main reference model
TopologyHistory	Topology plus compact history features	History ablation

2.2 PHYSICAL OCCUPANCY AND LOGICAL TOPOLOGY

Let B_t denote physical point occupancy and let $E_t^{(b)}, E_t^{(w)}$ denote the explicit logical edge sets of the two players. Physical line points are used for rendering and blocking, whereas logical edges define connectivity to the protected initial node. An attack can break an edge and remove nodes and line points that are no longer logically connected.

2.3 FULL SIMULATOR STATE

We model the full state as

$$x_t = \left(B_t, E_t^{(b)}, E_t^{(w)}, H_t, p_t, c_t, z_t \right), \quad (2)$$

where H_t is the visited-state set used by the repetition rule, p_t is the player to act, c_t is the consecutive-pass count, and z_t is the terminal flag and cached terminal outcome. Resignation is outside the frozen training environment.

2.4 OBSERVATION PROTOCOLS

All four protocols also expose the player to act, consecutive-pass count, and legal-action mask. Thus a legal-set difference alone is not an alias under the implemented Grid protocol; it is an alias only for a separately named mask-free occupancy projection.

2.5 RULES AND TERMINAL REWARD

The executable rules distinguish occupied nodes, physical line points, and logical edges. Point actions extend or attack connectivity subject to protected regions and the three-point restriction; attacks can cut edges and cascade through components disconnected from a player’s protected root. Superko transactionally rejects repeated canonical positions. Passing and automatic skips use the exact consecutive-pass terminal rule, after which discrete territory determines the winner. Training uses only terminal win/draw/loss reward $(+1, 0, -1)$.

2.6 IMPLEMENTATION AND VALIDATION

The dependency-free Python environment uses a fixed point ordering followed by PASS, lossless cloning and versioned serialization, legal-action masks, and replayable action sequences. The frozen environment/search suite contains 62 passing tests. An expanded repository discovery snapshot on 2026-08-25 ran 88 tests: 81 passed and seven optional PyTorch-dependent tests were skipped because the dependency was unavailable; skips are not counted as failures. Five targeted and 60 seeded random trajectories match the independent Web rule engine step by step across side lengths 5, 7, 9, 10, 12, and 15. Targeted tests assert exact component deletion, edge-only cuts, edge restoration, and transactional Superko rollback. Median end-to-end throughput in a three-repeat local engineering benchmark ranges from 1,574 transitions/s at side length 5 to 19.3 transitions/s at side length 15. These measurements include legal-mask and observation construction and are not optimized simulator-only timings.

3 STATE-ALIASING DIAGNOSTICS

Definition 1 (Implemented Grid equivalence). *Two full states x and x' are Grid-equivalent if their implemented Grid observations are identical: physical occupancy, fixed coordinate features, player to act, consecutive-pass count, and legal-action mask all agree. Logical edges and exact repetition histories may differ.*

We use *occupancy projection* for the deliberately weaker map that omits the action mask. Results for that projection are never reported as aliases of the implemented Grid observation.

Definition 2 (Decision-relevant aliasing). *An observation map ϕ aliases two states in a decision-relevant manner if $\phi(x) = \phi(x')$ but the states differ in legal actions, observable transition outcomes, returns, or optimal actions under a fixed game solution concept.*

Proposition 1 (Reachable topology aliasing: computational witness). *For side length $n = 5$, there exist two legal histories that reach states with identical implemented Grid observations but different logical edge sets.*

Proposition 2 (Side-five negative result: exhaustive computation). *Under the frozen two-player rules and standard side-five start, no pair of reachable Grid-equivalent states differs in legal actions, next Grid observation, exact zero-sum value, shared-action Q value, or optimal-action set. No reachable side-five move is rejected by Superko.*

Evidence and boundary. The complete raw graph contains 25,096 states and 67,505 transitions and is a DAG. It contains 976 Grid-alias groups and 1,088 pairs with different latent logical topology. An independent all-PASS backward-induction audit finds zero strict differences across every pair. This proves a side-five negative result, not nonexistence at sizes 6–15. The replay witness therefore establishes weak full-state aliasing only; the paper does not currently claim that Grid is non-Markov or strategically insufficient across the benchmark family.

Separately, a standard-start side-six trajectory naturally reaches a move that is rejected only by Superko. If enforcement is disabled while repetitions are still logged, that move closes a six-transition loop that can repeat. This establishes that the rule can bind, but not the stricter paired-state claim: we have not found two naturally reachable histories with identical mask-free Topology and different legality for a shared action.

Witness construction. Complete enumeration is combined with seeded random walks, collision indexing, and bounded shared-continuation search. Versioned witnesses retain legal histories, observation digests, full-state fingerprints, and replay assertions; unsuccessful larger-board searches retain their budgets and are labelled NOT_FOUND_WITHIN_BUDGET.

4 BENCHMARK PROTOCOL

4.1 SEARCH BASELINES

Random, Greedy, depth-2/3 Minimax, and UCT-MCTS share one color-balanced, replayable arena. Minimax retains at most 20 ordered point candidates plus PASS and is therefore approximate. The complete five-agent matrix remains an engineering smoke test; formal neural evaluation uses Random and UCT-MCTS-16.

4.2 NEURAL SELF-PLAY

All models use the same AlphaZero-style pipeline: 16 PUCT simulations, $c_{\text{puct}} = 1.5$, legal-masked priors, root Dirichlet noise $(\alpha, \epsilon) = (0.3, 0.25)$, and temperature 1.0 for 20 plies then 0.1. Each model-seed run has 100 complete mixed-size games and 200 AdamW updates. Complete games contribute root-visit policies and terminal outcomes from the stored player-to-act perspective; truncations never enter replay. Checkpoints at steps 40, 80, 120, 160, and 200 store the optimizer, replay, counters, and RNG states and resume only after hash validation.

Table 2: Matched model families used in the formal comparison.

Model	Width	Parameters	Structural input
Padded-CNN	33	62,868	node features only
Grid-GNN	82	62,733	physical lattice edges
Topology-GNN	64	63,171	physical + typed logical edges

4.3 COMPUTE MATCHING AND STATISTICS

The comparison matches five seeds, training schedule, search, optimizer, checkpoints, evaluation games, and parameter count (maximum spread 0.6982%). Wall-clock, transitions, FLOPs, and memory are reported but not matched. Each final seed-level matchup contains 100 same-seed color-swapped pairs. Pooled Wilson intervals count draws as half a success and are descriptive: the deterministic round robin contains only two action sequences per 200-game task, so no independent-game or multiple-comparison inference is claimed.

5 TOPOLOGY-AWARE REFERENCE MODEL

Each lattice point is a graph node with occupancy, role, coordinate, and initial-node features. Typed edges represent (i) fixed triangular-lattice adjacency, (ii) black logical connections, and (iii) white logical connections. A relation-aware message-passing network produces per-node action logits and a pooled scalar value. A separate global logit represents pass.

All inputs are canonicalized to the player to act. Each node receives 11 features: five occupancy indicators, two normalized coordinates, boundary membership, own and opponent initial-node indicators, and the consecutive-pass count. For relation $r \in \{\text{physical, own, opponent}\}$, one message-passing layer has the form

$$h_i^{(\ell+1)} = \rho \left(W_0^{(\ell)} h_i^{(\ell)} + \sum_r W_r^{(\ell)} \frac{1}{|\mathcal{N}_r(i)|} \sum_{j \in \mathcal{N}_r(i)} h_j^{(\ell)} \right), \quad (3)$$

with a residual connection and masked pooling over valid nodes. Grid-GNN exposes only the physical relation; Topology-GNN additionally remaps the two color-specific logical edge sets to own/opponent relations. Padded-CNN maps the same node features to a zero-padded square tensor and does not read any edge relation. All models emit one logit per point, a separate PASS logit, and a $\tanh$ value; padding and illegal actions are masked.

Each network uses three residual convolution/message-passing blocks. Widths were chosen before the formal run to keep the maximum parameter spread below the frozen 1% tolerance. TopologyHistory remains a declared observation variant but was not included in the neural comparison.

6 EXPERIMENTS

6.1 ENVIRONMENT AND DIAGNOSTIC EVIDENCE

The frozen environment/search suite has 62 passing tests; 65 targeted or seeded trajectories match an independent Web implementation over sizes 5–15. Complete side-five enumeration covers 25,096 states and 67,505 transitions. All 1,088 topology-distinct Grid-equivalent pairs have identical legal sets, one-step observations, exact values, shared-action values, and optimal actions. A PASS-aware side-six search finds a natural Superko trigger, while a separate 5,000-pair, 12,656-shared-action bounded continuation search finds no strict paired witness. The latter is a budgeted miss, not an absence proof.

6.2 WITHIN-SIZE SELF-PLAY

The formal ledger contains 15 training and 315 evaluation tasks: 1,500 self-play games, 3,000 updates, and 30,600 unique replay-verified arena games, with no failure or truncation. All nine model–

Table 3: Final-checkpoint score rates on the three training sizes. Each cell pools 1,000 games over five training seeds; brackets are descriptive Wilson 95% intervals.

Model	n	vs. Random	vs. UCT-MCTS-16
Grid-GNN	5	.851 [.827,.871]	.523 [.492,.553]
	7	.892 [.871,.909]	.646 [.615,.675]
	9	.951 [.936,.963]	.848 [.824,.869]
Padded-CNN	5	.873 [.851,.892]	.532 [.501,.562]
	7	.939 [.922,.952]	.711 [.682,.738]
	9	.978 [.967,.985]	.914 [.895,.929]
Topology-GNN	5	.863 [.840,.883]	.643 [.612,.672]
	7	.910 [.890,.926]	.715 [.686,.742]
	9	.949 [.933,.961]	.876 [.854,.895]

size cells beat Random (Table 3). Against UCT-MCTS-16, Grid-GNN at $n = 5$ is inconclusive, Padded-CNN at $n = 5$ is borderline, and the other seven cells have descriptive lower bounds above 0.5.

Five checkpoint probes show non-monotone, size-dependent trajectories. Because no area-under-curve or fixed-strength estimator was preregistered, they do not establish relative sample efficiency.

6.3 REPRESENTATION ROUND ROBIN

Direct play exhibits a size–representation interaction (Table 4). Topology-GNN outscores Grid-GNN at all three sizes, while Padded-CNN reverses direction against both graph models. No global ordering is supported.

Table 4: Score from A’s perspective over five matched training seeds. Intervals are omitted because trajectories repeat deterministically.

A vs. B	$n = 5$	$n = 7$	$n = 9$
Topology-GNN vs. Grid-GNN	.65	.80	.60
Padded-CNN vs. Grid-GNN	.50	.55	.30
Padded-CNN vs. Topology-GNN	.60	.30	.50

Across all 9,000 round-robin games, each 200-game task contains exactly two unique color-conditioned trajectories. Each aggregate therefore represents five training seeds and ten unique trajectories, not 1,000 independent trials.

6.4 ABLATIONS

`enforce` rejects repetitions, whereas `observe` admits them while retaining history. In a 2,000-pair fixed-policy pilot at each of sizes 6 and 7, only two size-7 games triggered Superko and one trajectory diverged; all winners agreed, no game truncated, and mean length changed by -0.01017% .

All internal fixed gates for score (± 5 percentage points), relative length ($\pm 5\%$), and truncation passed for this policy, size set, and horizon. This supports “empirically similar under the sampled policy,” not rule equivalence: only one handcrafted policy was evaluated.

A preregistered five-seed extension covered six policies, two sizes, 120 matched pairs, and 240 replay-verified games. It observed no trigger or outcome effect, but every practical-similarity gate failed: each cell had only ten `observe` games, so 0/10 has a 27.75% Wilson upper bound, above the 1% threshold. CycleSeeking’s null result is a bounded search miss, and its neural row is a one-game, one-update smoke checkpoint.

6.5 COMPUTE AND REPRODUCIBILITY

Runs used Python 3.10.20, PyTorch 2.7.1+cu118, deterministic CUDA 11.8, and an RTX 4060 Laptop GPU. Mean per-seed training time was 489.8/399.2/440.0 s for Padded-CNN/Grid-GNN/Topology-

GNN (1.85 aggregate training GPU-hours); evaluation compute is not yet consolidated. One canonical manifest digest (1bbf2c9a9ecc . . .) binds all 330 receipts. Independent aggregation rechecked task identity, budgets, hashes, summaries, and 30,600 replays; `formal_ready=true` denotes artifact completeness, not superiority.

7 LIMITATIONS AND BROADER IMPACT

- Results cover jointly trained sizes 5, 7, and 9, not zero-shot sizes 10 or 12; history, logical-edge, cascade, and search-budget ablations remain.
- CNN/GNN comparisons mix representation and architecture. Parameters and data are matched, but FLOPs, memory, transitions, and wall time are not.
- Deterministic round-robin trajectories invalidate independent-game uncertainty; diversified evaluation and seed-aware inference are required.
- LIFELINE-RL is hand-designed, and its Superko matrix is finite-policy evidence. The smoke neural condition and incomplete evaluation-compute accounting preclude broader efficiency claims.

8 CONCLUSION

LIFELINE-RL isolates dynamic topology and history-dependent legality in a replayable benchmark. Side-five enumeration yields a decision-relevant negative result, while a side-six trace proves Superko can bind. Under one frozen five-seed protocol, all three learned representations beat Random on seen sizes; Topology-GNN outscores Grid-GNN directly, but size-dependent reversals and duplicated trajectories rule out a global ranking. Next steps are diversified seed-aware evaluation, zero-shot sizes, and isolated rule/feature ablations.

A HYPERPARAMETERS AND TRAINING DETAILS

Table 5: Frozen formal training and evaluation settings shared by all three learned representations.

Setting	Value
Training sizes / seeds	{5, 7, 9} / 20260825–20260829
PUCT	16 simulations, $c_{\text{puct}} = 1.5$
Root noise	$\alpha = 0.3$, $\epsilon = 0.25$
Temperature	1.0 for 20 plies, then 0.1
Self-play / updates	100 complete games / 200 gradient steps
Replay / batch	10,000 / 16
Optimizer	AdamW, lr 10^{-3} , weight decay 10^{-4}
Gradient clipping	global norm 5.0
Checkpoints	steps 40, 80, 120, 160, 200
Terminal rule	256 plies; enforced Superko; no shaped reward
Final evaluation	200 games per seed and matchup
Color control	100 same-seed swapped-color pairs

Targets are the masked root-visit distribution and the terminal result from the stored player-to-act perspective. Absolute player payoffs are backed up through the search tree because automatic skips can leave the same player to act after a transition. Training loss is masked policy cross-entropy plus mean-squared terminal value error.

B ADDITIONAL RESULTS

Canonical aggregate tables retain all checkpoints, W/D/L, seed identities, intervals, losses, transitions, and wall time. The formal run has no failed or truncated task; smoke runs and duplicate views of the step-200 UCT matchup are excluded from the 30,600 unique games.

C REPRODUCIBILITY CHECKLIST

The release includes exact source, frozen manifest, parameter audit, task bundle, append-only ledger, checkpoints, JSONL logs, CSV summaries, hashes, and independent verifiers. Aggregation fails on missing/duplicate tasks, identity or budget drift, truncation, color imbalance, or incomplete replay. Two interrupted runner invocations are documented; their first two attempts lack independent canonical attempt files, so per-attempt lineage is not claimed.